

4.1.8 Emergency stop

The emergency stop is always installed according to local safety regulations. It is often a button placed on the wall near the separator or on the control equipment.

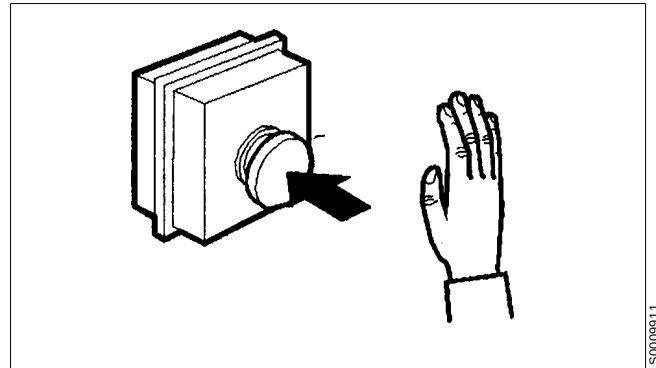
The following should happen:

- The bowl is kept filled with liquid until standstill.
- The separator motor is switched off.

Evacuate the room. The separator may be hazardous when passing resonance frequencies during the run-down.

After an emergency stop the cause of the fault must be identified.

If all parts have been checked and the cause remains unclear contact your Alfa Laval representative for advice.



Emergency stop

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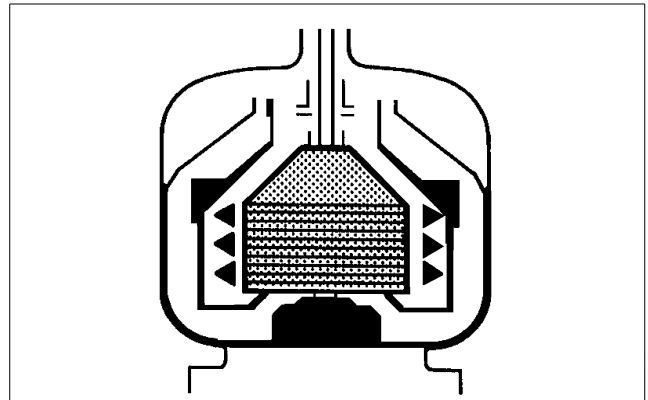
4.2 Cleaning the bowl

The separated sludge is accumulating on the inside surface of the separator bowl. How often the separator needs to be cleaned, depends on the amount of sediment entering the separator. High solids content or high throughput has the consequence that the cleaning need to be done more often.

Intervals for a specific case must be based on experience.

Information about sludge content in the feed can be achieved either by using a lab centrifuge or to run the separator for 2 - 3 hours and then stop and open the bowl to check the amount of sludge.

Then prolong or shorten the time between stop and cleaning.



Sludge accumulation

4.2.1 Removal of separated sludge

Remove the sludge collected on the inside of the bowl as follows:

1. Stop the separator.

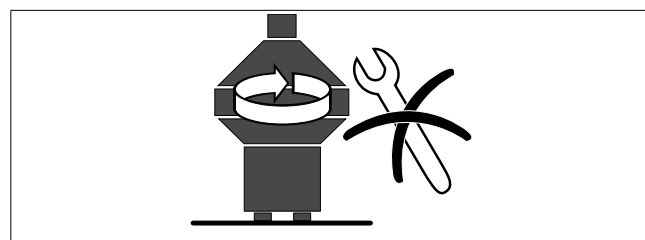


DANGER

Entrapment hazards

Make sure that rotating parts have come to a **complete standstill** before starting **any** dismantling work.

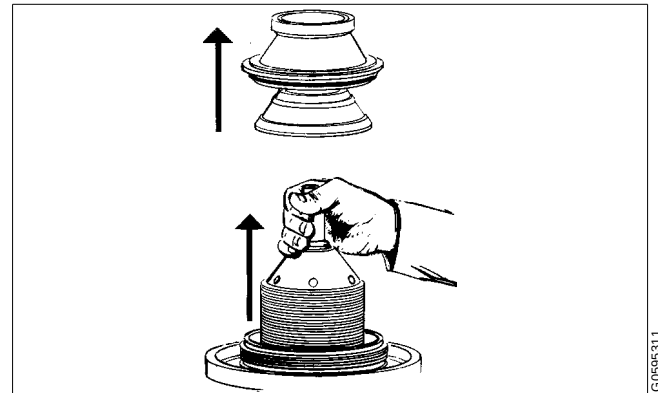
The revolution counter and the motor fan indicate if the separator is rotating or not.



2. Loosen the hinged bolts. Remove the collecting cover.

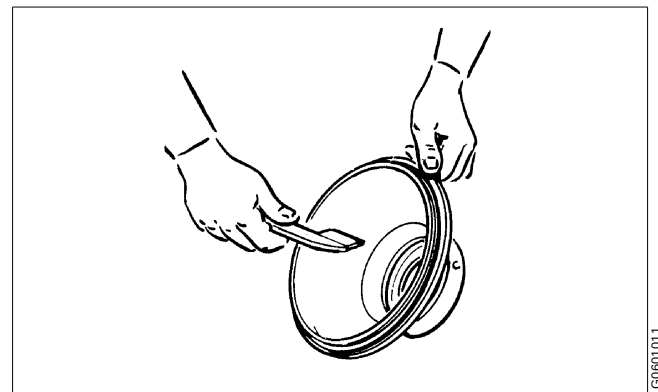
For instructions, see chapter "6 Dismantling/Assembly" on page 83.

3. Open the separator bowl.



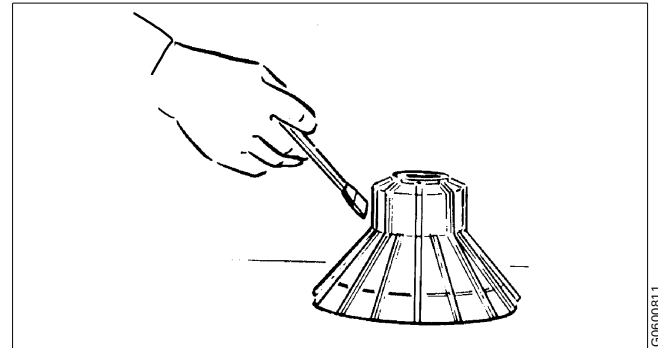
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4. Clean the bowl hood.



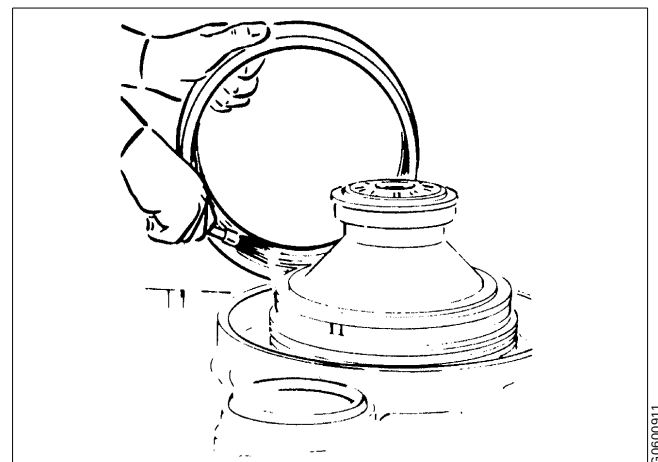
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5. Clean the channels on the top disc upper side.



G0600811

6. Remove sediment from the bowl body, clean and lubricate lock ring.



G0600911

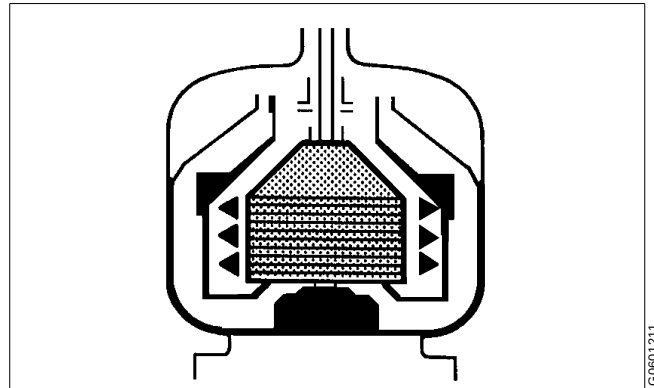
4.2.2 Disc stack

When the sediment is not sticky, the disc stack can be cleaned by "hurling".

1. Clean the other bowl parts.
2. Assemble the bowl.
3. Close and lock the collecting cover.
4. Run up to full speed without liquid feed.
5. After "hurling" either continue separation or stop and open the bowl and remove the sediment.

If the sediment adheres firmly, dissolve it by submerging the distributor and the disc stack in a suitable detergent.

If "hurling" has no effect, clean the discs one by one.



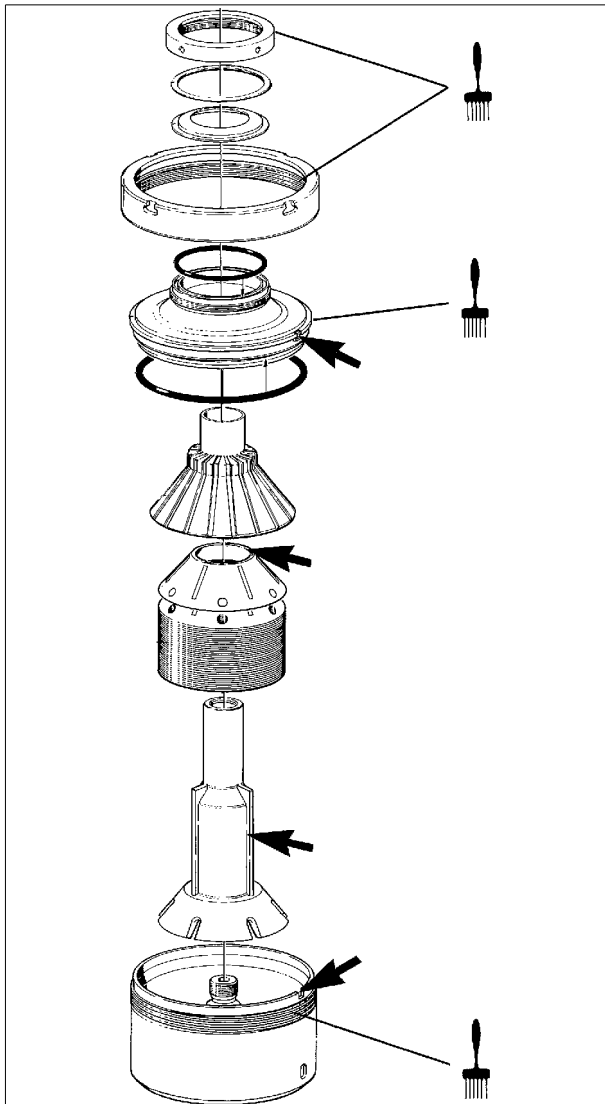
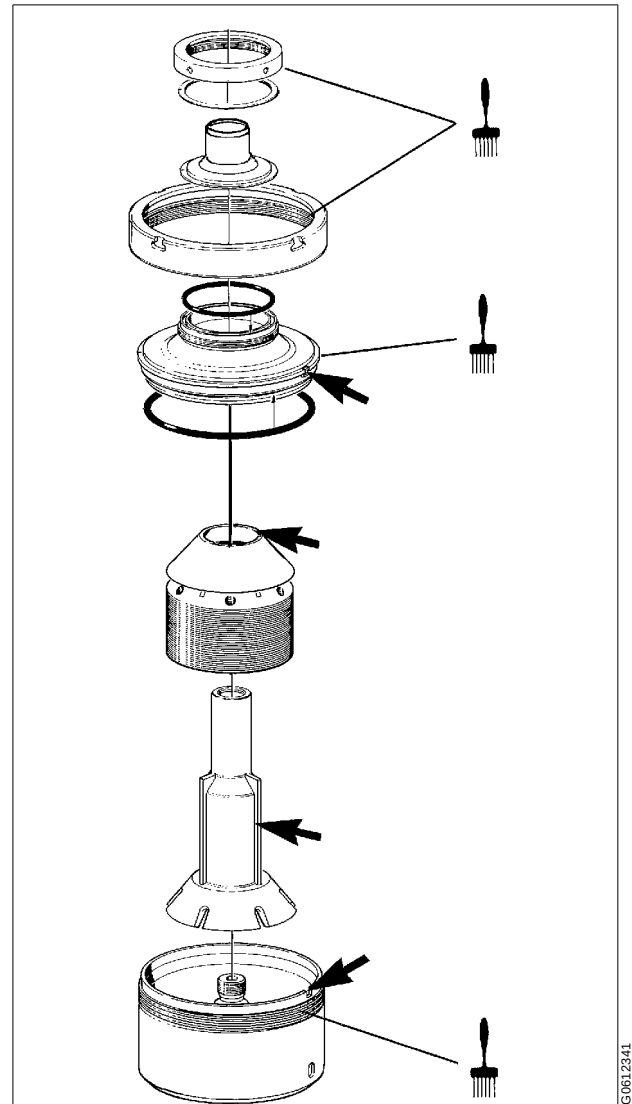
Bowl cleaning by "hurling"

4.2.3 Assembling the bowl

Each bowl constitutes a balanced unit. Exchange of any major part may need rebalancing of the bowl. To prevent mixing of parts, e.g. in an installation comprising of several machines of the same type, the major bowl parts carry the machine manufacturing number or its last three digits.

NOTE

Be sure bowl parts are not interchanged. Out of balance vibration will reduce ball bearing life.

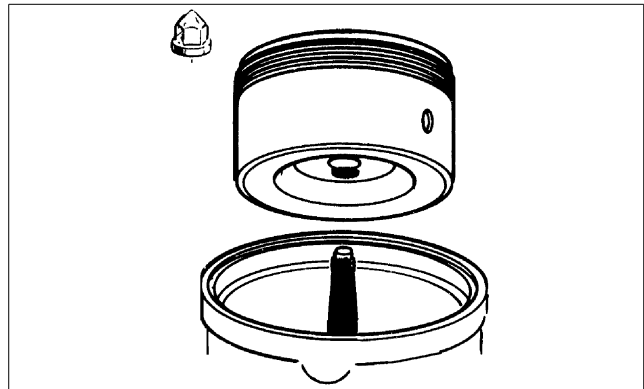
Purifier and clarifier bowl*Purifier bowl**Clarifier bowl*

The arrows indicate position of guides in the bowl.

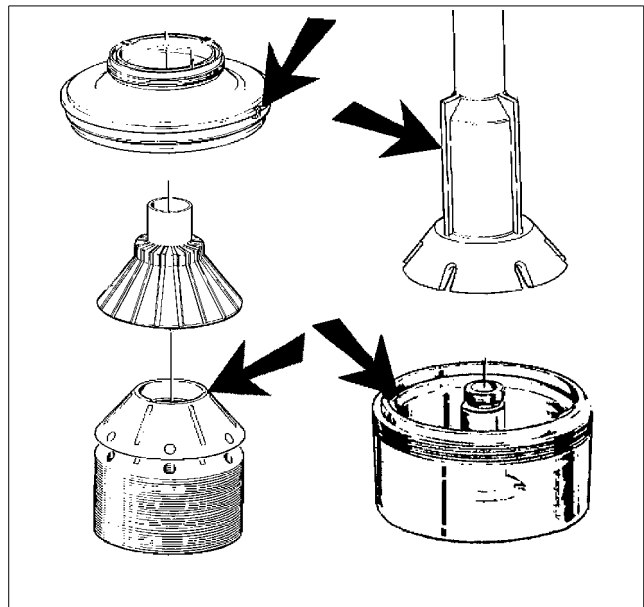
*Lubrication needed
Molykote 1000*



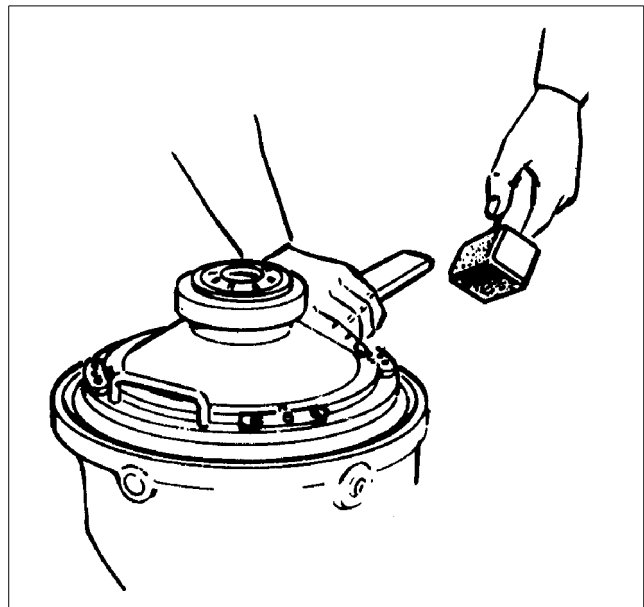
1. Clean spindle top and bowl body nave with a cloth.



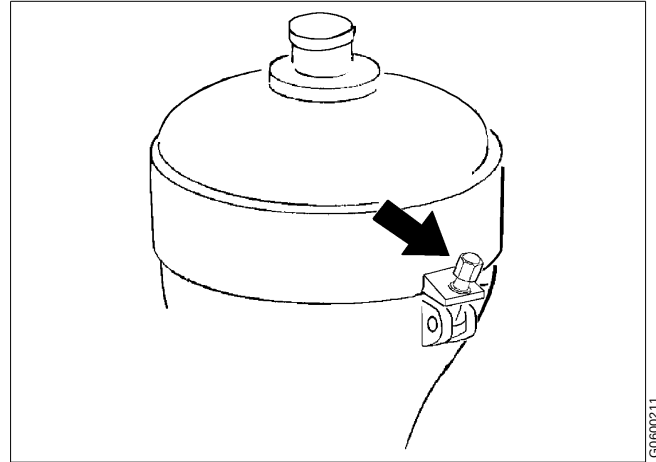
2. Bring bowl parts into positions defined by the guides.



3. Screw large lock ring anti-clockwise until bowl hood lies tightly against bowl body.



4. Assemble and clamp the collecting cover.
Tighten the hook screws.



Hook bolts on collecting cover

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5 *Service Instructions*

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5.1 Periodic maintenance

5.1.1 Introduction

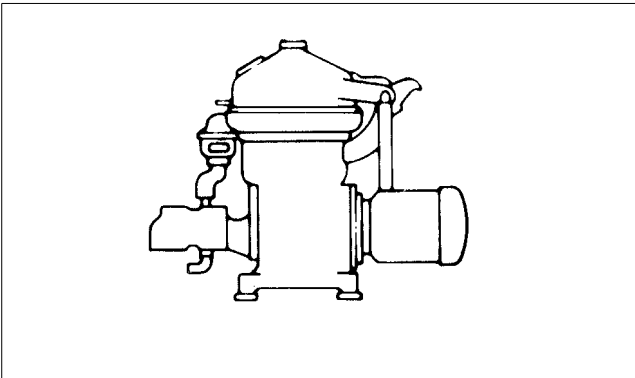
Periodic, preventive maintenance reduces the risk of unexpected stoppages and breakdowns. Maintenance logs are shown on the following pages in order to facilitate periodic maintenance.



DANGER

Disintegration hazards

Separator parts that are worn beyond their safe limits or incorrectly assembled may cause severe damage or fatal injury.



Periodic maintenance prevent stoppages

5.1.2 Maintenance intervals

The following directions for periodic maintenance give a brief description of which parts to clean, check and renew at different maintenance intervals.

The Service Logs for each maintenance interval later in this chapter give detailed enumeration of the checks that must be done.

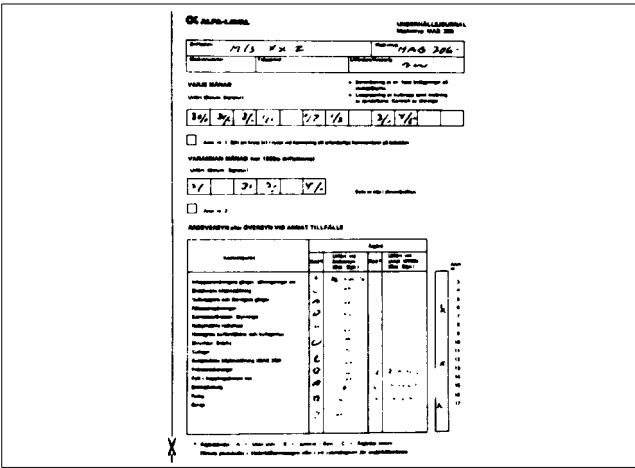
Daily checks consist of simple check points to carry out for detecting abnormal operating conditions.

Oil change interval is every 1500 hours or at least once every year if the total number of operating hours is less than 1500 hours.

Time of operation between oil changes can be extended from the normal 1500 hours to 2000 hours if a synthetic oil of group D is used.

In seasonal operation change the oil before a new period.

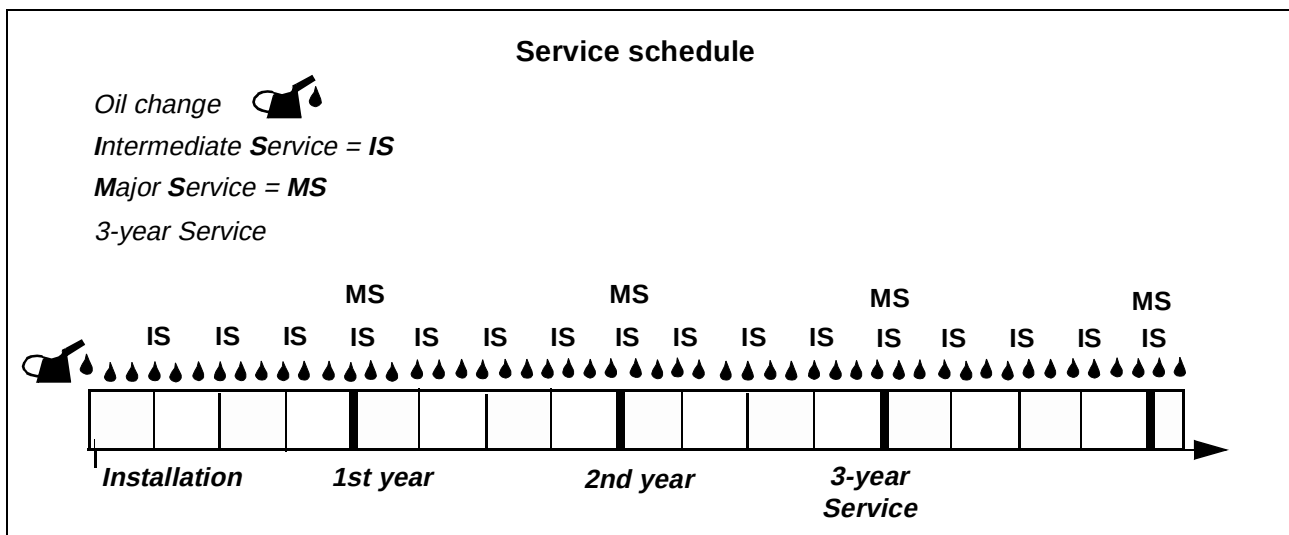
IS - Intermediate Service consists of an overhaul of the separator bowl, inlet and outlet every 3 months or 2000 operating hours. Seals in bowl and gaskets in the inlet/outlet device are renewed.



Maintenance log

MS - Major Service consists of an overhaul of the complete separator every 12 months or 8000 operating hours. An Intermediate Service is performed, and the flat belt, friction elements, seals and bearings in the bottom part are renewed.

3-year service consists of service of the coupling bearings, service of frame intermediate part and renewal of frame feet. The rubber feet get harder with increased use and age.



Other

Check and prelubricate spindle bearings of separators which have been out of service for 6 months or longer.

NOTE

Do not interchange bowl parts!

To prevent mixing of parts, e.g. in an installation comprising several machines of the same type, the major bowl parts carry the machine manufacturing number or its last three digits.

5.1.3 Maintenance procedure

At each Intermediate and Major Service, take a copy of the Service Log and use it for notations during the service.

An Intermediate and Major Service should be carried out in the following manner:

1. Dismantle the parts as mentioned in the Service Log and described in Chapter "6 Dismantling/Assembly" on page 83.

Place the separator parts on clean, soft surfaces such as pallets.
2. Inspect and clean the dismantled separator parts according to the Service Log.
3. Fit all the parts delivered in the Service kit while assembling the separator as described in chapter "6 Dismantling/Assembly" on page 83. The assembly instructions have references to check points which should be carried out during the assembly.

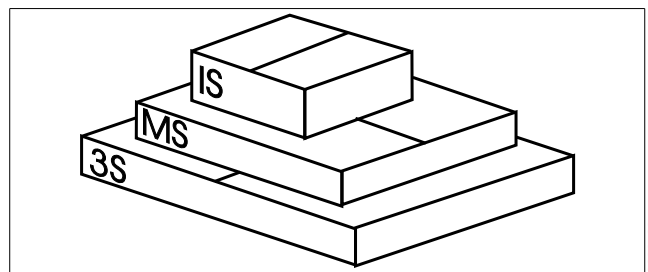
5.1.4 Service kits

Special service kits are available for Intermediate Service (IS), Major Service (MS) and 3-year service (3-S).

For other services the spare parts have to be ordered separately.

Note that the parts for IS are **not** included in the MS kit.

The contents of the service kits are described in the *Spare Parts Catalogue*.



Spare parts kit

NOTE

Always use Alfa Laval genuine parts as otherwise the warranty will become invalid.

Alfa Laval takes no responsibility for the safe operation of the equipment if non-genuine spare parts are used.

5.2 Maintenance Logs

Keep a log of inspection and maintenance performed. Parts repeatedly replaced should be given special consideration. The cause of repeated failures should be determined and corrected. Discuss your problems with an Alfa Laval representative and, when necessary, request a visit by an Alfa Laval Service engineer.

Rate of corrosion and erosion and notification of cracks should also be a part of this log. Note the extent of damage and date the log entries so that the rate of deterioration can be observed.

5.2.1 Daily checks

The following steps should be carried out daily.

Main component and activity	Part	Page	Notes
Inlet and outlet			
Check for leakage	Collecting cover	-	
Separator bowl			
Check for vibration and noise		77	
Worm wheel shaft and gear casing			
Check for vibration and noise			
Check	Oil level in gear housing	54	
Electrical motor			
Check for heat, vibration and noise		_1)	

¹⁾ See manufacturer's instruction

5.2.2 Oil change

The oil change and check of worm gear should be carried out every 1500 * hours of operation.

Main component and activity	Part	Page	Notes
Worm wheel shaft and gear housing			
Check	Worm wheel and worm	69	
Renew	Oil * in gear housing	54	

* When using a group D oil, time of operation between oil changes can be extended from the normal 1500 hours to 2000 hours.

When the separator is running for short periods, the lubricating oil must be changed every 12 months even if the total number of operating hours is less than 1500 hours (2000 h).

In a new installation, or after replacement of gear, change the oil after 200 operating hours.

See chapter "8 Technical Reference" on page 125 for further information.

5.2.3 IS - Intermediate Service

Name of plant:

Local identification:

Separator: MAB 104B-14/24C

Manufacture No./Year:

Total running hours:

Product No.: 881241-08-16

Date:

Signature:

Renew all parts included in the Intermediate Service kit (IS) and do the following activities.

Main component and activity	Part	Page	Notes
Inlet and outlet, frame			
Clean and inspect	Housings and frame hood	—	
Separator bowl			
Clean and inspect	Bowl hood	—	
	Top disc	—	
	Bowl discs	75	
	Distributor		
	Threads on lock ring and bowl body	63	
Check	Galling of guide surface	63	
	Corrosion, erosion, cracks	58 - 61	
Power transmission			
Change	Oil in oil sump	76	
Electrical motor			
Lubrication (if nipples are fitted)	See sign on motor		
Signs and labels on separator			
Check attachment and legibility	Safety label on hood	158	
	Other plates and labels		

5.2.4 MS-Major Service

Name of plant:

Local identification:

Separator: MAB 104B-14/24C

Manufacture No./Year:

Total running hours:

Product No.: 881241-08-16

Date:

Signature:

Main component and activity	Part	Page	Notes
Inlet and outlet			
Clean and inspect	Threads of inlet	–	
Separator bowl			
Clean and check	Lock ring	63	
	Bowl hood	–	
	Top disc	–	
	Gravity disc	–	
	Bowl discs	75	
	Distributor	–	
	Bowl body	–	
	Corrosion	58	
	Erosion	60	
	Cracks	61	
	Disc stack pressure	62	
Renew	O-rings and sealings	–	
Worm wheel shaft and gear housing			
Check	Worm wheel and worm	69	
	Radial wobble of worm wheel shaft	-	
	Axial play of coupling disc	67	
Renew	Oil in gear housing	54	

Main component and activity	Part	Page	Notes
Vertical driving device			
Clean and check	Bowl spindle	66	
	Buffers		
	Ball bearing housing indentations	64	
	Radial wobble of bowl spindle	64	
Renew	Ball bearings and top bearing springs	66	
Frame feet			
Renew	Rubber cushions	114	
Electrical motor			
Clean and check	Position of coupling disc	67	
Lubrication (if nipples are fitted)	–	–	
Signs and labels on separator			
Check attachment and readability, replace if needed	Safety label on collecting cover	158	
Monitoring equipment (option)			
Function check	Cover interlocking switch	71	

5.3 MS - Check points

5.3.1 Corrosion

Evidence of corrosion attacks should be looked for and rectified each time the separator is dismantled. Main bowl parts such as the bowl body, bowl hood and lock ring must be inspected with particular care for corrosion damage.



DANGER

Disintegration hazard

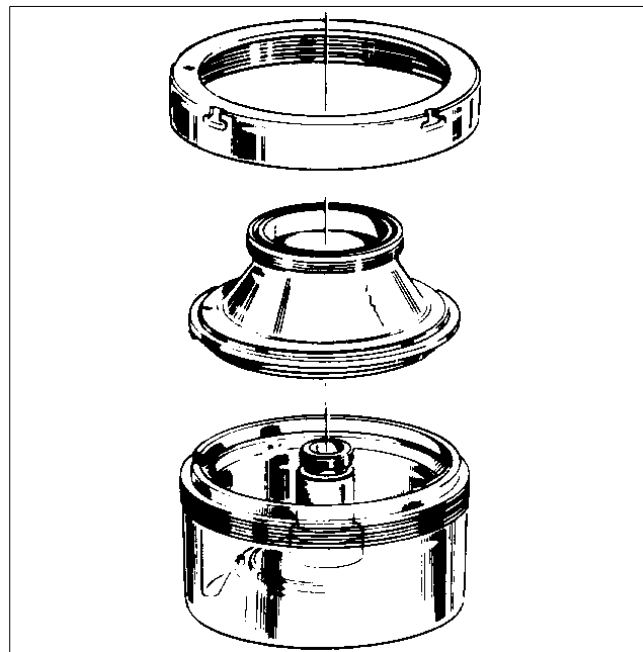
Inspect regularly for corrosion damage.
Inspect frequently if the process liquid is corrosive.

Always contact your Alfa Laval representative if you suspect that the largest depth of the corrosion damage exceeds 1,0 mm or if cracks have been found. Do not continue to use the separator until it has been inspected and given clearance for operation by Alfa Laval.

Cracks or damage forming a line should be considered as being particularly hazardous.

Non-stainless steel and cast iron parts

Corrosion (rusting) can occur on unprotected surfaces of non-stainless steel and cast iron. Frame parts can corrode when exposed to an aggressive environment.



Main bowl parts

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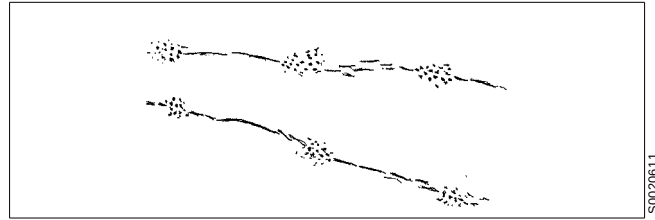
Stainless steel

Stainless steel parts corrode when in contact with either chlorides or acidic solutions. Acidic solutions causes a general corrosion. The chloride corrosion is characterised by local damage such as pitting, grooves or cracks. The risk of chloride corrosion is higher if the surface is:

- Exposed to a stationary solution.
- In a crevice.
- Covered by deposits.
- Exposed to a solution that has a low pH value.

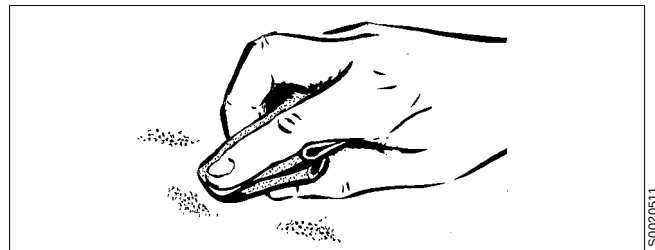
Corrosion damage caused by chlorides on stainless steel begins as small dark spots that can be difficult to detect.

- Inspect closely for all types of damage by corrosion and record these observations carefully.
- Polish dark-coloured spots and other corrosion marks with a fine grain emery cloth. This may prevent further damage.



Corrosion forming a line

S0020611



Polish corrosion spots

S0020511



DANGER

Disintegration hazard

Pits and spots forming a line may indicate cracks beneath the surface.
All forms of cracks are a potential danger and are totally unacceptable.
Replace the part if corrosion can be suspected of affecting its strength or function.

Other metal parts

Separator parts made of materials other than steel, such as brass or other copper alloys, can also be damaged by corrosion when exposed to an aggressive environment. Possible corrosion damage can be in the form of pits and/or cracks.

5.3.2 Erosion

Erosion can occur when particles suspended in the process liquid slide along or strike against a surface. Erosion can become intensified locally by flows of higher velocity.



DANGER

Disintegration hazard

Inspect regularly for erosion damage. Inspect frequently if the process liquid is erosive.

Always contact your Alfa Laval representative if the largest depth of any erosion damage exceeds 1,0 mm. Valuable information as to the nature of the damage can be recorded using photographs, plaster impressions or hammered-in lead.

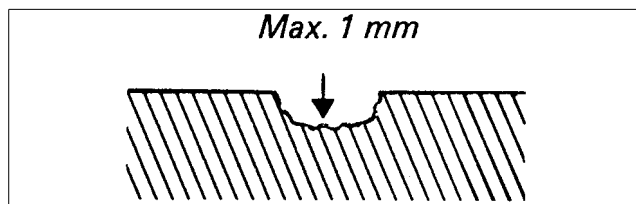
Erosion is characterised by:

- Burnished traces in the material.
- Dents and pits having a granular and shiny surface.

Surfaces particularly subjected to erosion are:

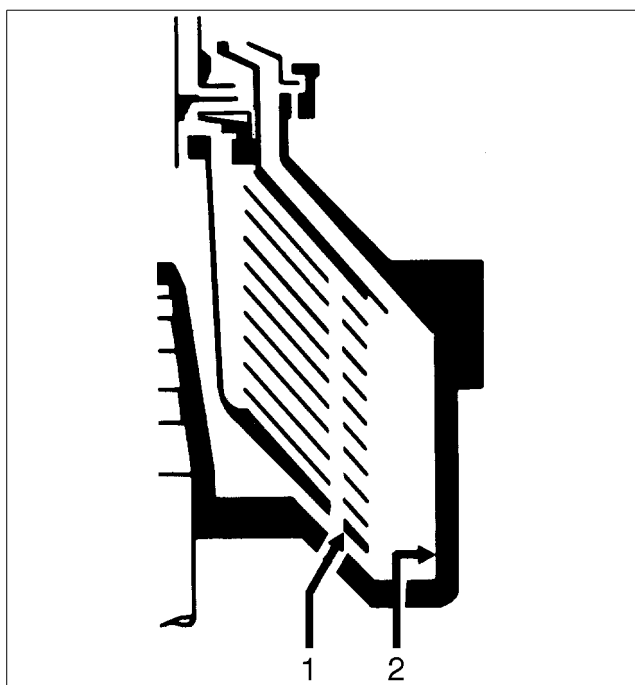
1. The underside of the distributor in the vicinity of the distribution holes and wings.
2. The internal surface of the bowl body that faces the conical part of the distributor.

Look carefully for any signs of erosion damage. Erosion damage can deepen rapidly and consequently weaken parts by reducing the thickness of the metal.



Maximum depth of damage

G0206221



Erosion check points

G0689911

**DANGER**
Disintegration hazard

Erosion damage can weaken parts by reducing the thickness of the metal.

Replace the part if erosion can be suspected of affecting its strength or function.

5.3.3 Cracks

Cracks can initiate on the machine after a period of operation and propagate with time.

- Cracks often initiate in an area exposed to high cyclic material stresses. These are called fatigue cracks.
- Cracks can also initiate due to corrosion in an aggressive environment.
- Although very unlikely, cracks may also occur due to the low temperature embrittlement of certain materials.

The combination of an aggressive environment and cyclic stresses will speed-up the formation of cracks. Keeping the machine and its parts clean and free from deposits will help to prevent corrosion attacks.

**DANGER**
Disintegration hazard

All forms of cracks are potentially dangerous as they reduce the strength and functional ability of components.

Always replace a part if cracks are present.

It is particularly important to inspect for cracks in rotating parts.

Always contact your Alfa Laval representative if you suspect that the largest depth of the damage exceeds 1,0 mm. Do not continue to use the separator until it has been inspected and cleared for operation by Alfa Laval.

5.3.4 Disc stack pressure

The lock ring (1) should press the bowl hood (2) firmly against the bowl body (3). The hood in turn should exert a pressure on the disc stack (4), clamping it in place.

NOTE

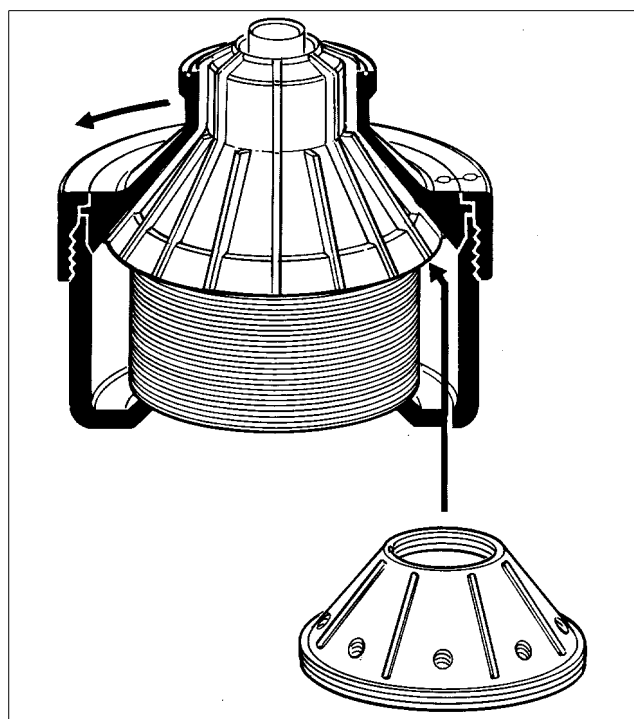
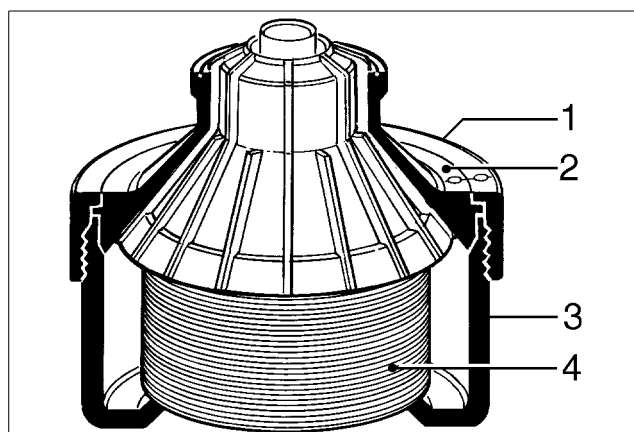
Insufficient pressure in disc stack can cause out of balance vibration and reduced lifetime of ball bearings.

Compress the disc stack by tightening the lock ring, see chapter "6.3.3 Bowl" on page 101.

Correct pressure is obtained when it is possible to tighten the lock ring so far by hand that the ϕ -mark on the lock ring is positioned 60° - 90° before the mark on the bowl hood.

To achieve this, add an appropriate number of discs to the top of the disc stack beneath the top disc.

Then advance the lock ring by giving the spanner handle some blows till the ϕ -marks are passed and the bowl is fully assembled.



5.3.5 Lock ring; wear and damage

Excessive wear or impact marks on threads, guide and contact surfaces of the lock ring, bowl hood and bowl body may cause hazardous galling.

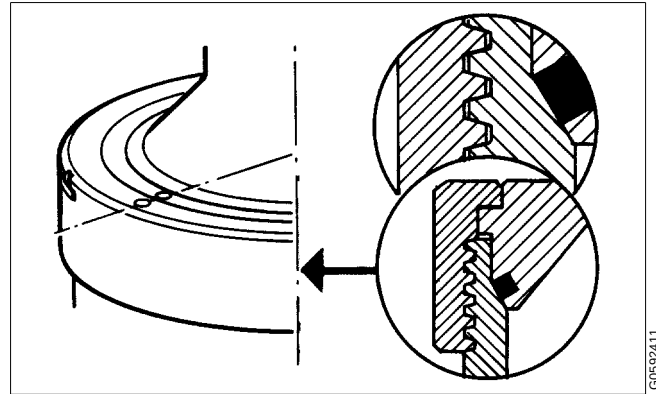
Check the thread condition by tightening the lock ring after removing the disc stack and bowl hood O-ring from the bowl.



DANGER

Disintegration hazards

Wear on large lock ring thread must not exceed safety limit. The ϕ -mark on lock ring must not pass opposite ϕ -mark by more than the specified distance.



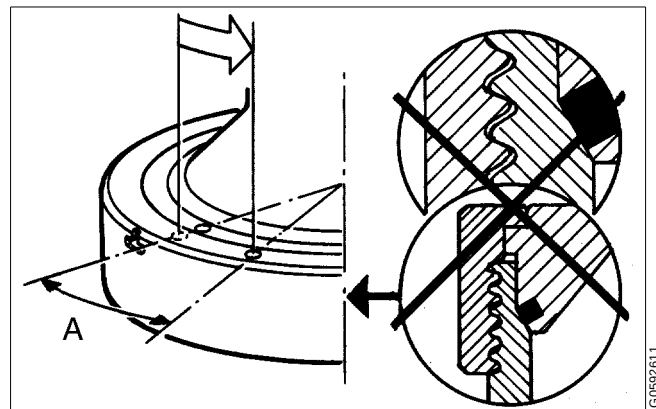
G0592411

In a new bowl the alignment marks on the lock ring and the bowl hood are exactly opposite each other.

If thread wear is observed, mark the bowl hood at the new position of the alignment mark on the lock ring by punching in a new alignment mark.

Contact Your Alfa Laval representative

- If the original mark on the lock ring passes the corresponding mark on the bowl hood by more than 25° which corresponds to **45 mm**.
- If the alignment marks become illegible. The thread wear need to be inspected and the new position of alignment marks determined.



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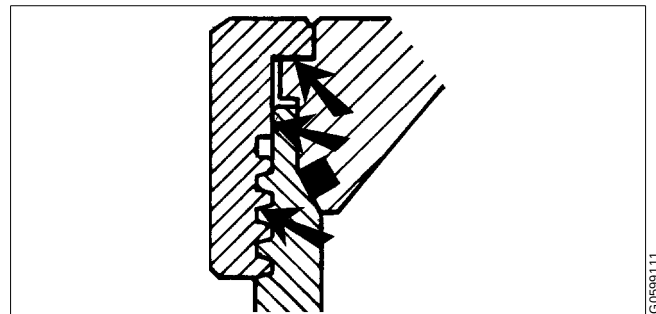
Maximum wear $A=25^\circ$

Damage

The position of the threads, contact and guide surfaces are indicated by arrows in the illustration.

Clean the threads, contact and guide surfaces with a suitable degreasing agent.

Check for burrs and protrusions caused by impact. Watch your fingers for sharp edges.



G0599111

If damage is established, rectify using a whetstone or fine emery cloth (recommended grain size 240).

If the damage is considerable, use a fine single-cut file, followed by a whetstone.

5.3.6 Radial wobble of bowl spindle

- Spindle wobble is indicated by rough bowl run (vibration).

The bowl spindle wobble should be checked if the bowl spindle has been dismantled or if rough bowl running (vibration) occurs.

NOTE

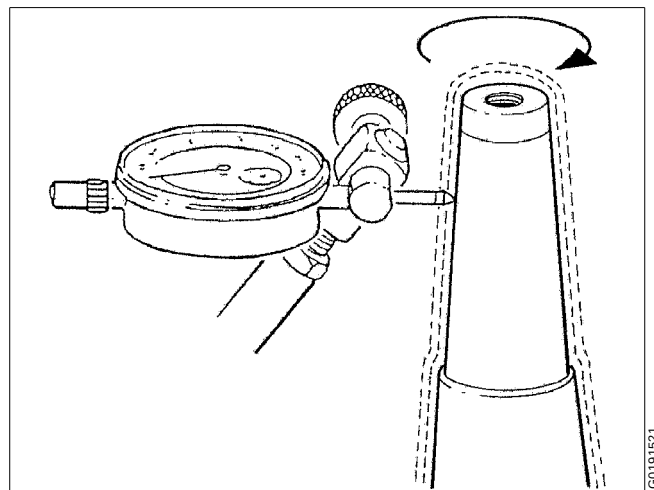
Spindle wobble will cause rough bowl run. This leads to vibration and reduces lifetime of ball bearings.

Check the wobble before mounting the bowl.

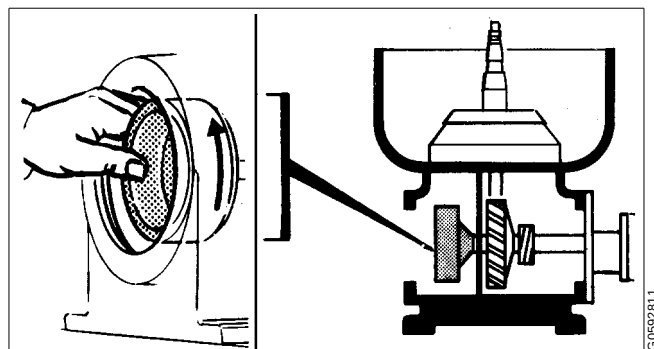
Before measuring, make sure that the buffer plugs are properly tightened.

- Fit a dial indicator in a support and fasten it to the frame.
- Remove the motor to get access to the coupling drum. Use the coupling drum to revolve the spindle manually.
- Measure the wobble at the top of the tapered end of the spindle. Maximum permissible radial wobble is **0,15 mm**.
- If wobble is too large, renew all the ball bearings on the spindle.

Measure wobble after assembly. If it is still excessive, the spindle is probably damaged and must be replaced, contact your Alfa Laval representative.



Maximum spindle wobble 0,15 mm



Rotate spindle by hand

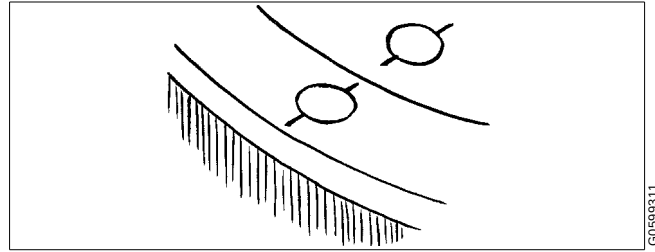
5.3.7 Height position of bowl / bowl spindle

Whenever the bowl spindle has been removed, its height position relative to the frame must be checked.

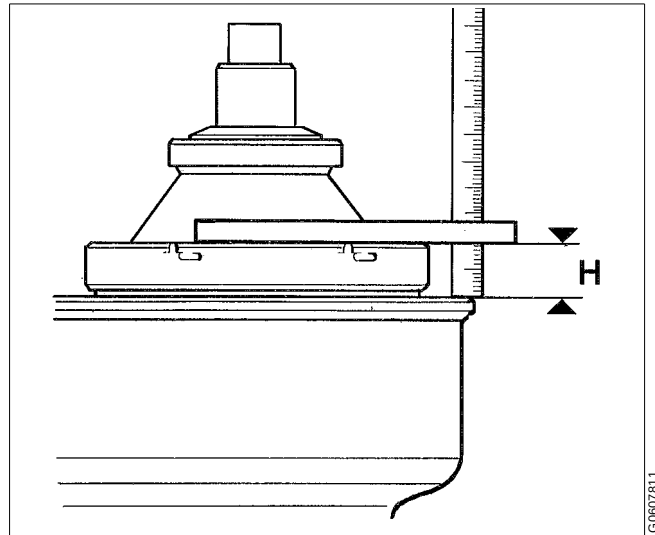
The bowl must be in place in the separator. The lock ring tightened. Make sure the alignment marks ϕ on bowl hood and lock ring are in front of each other.

For correct height position the distance (H) between the top face of the large lock ring and the top rim of frame must be **34-36 mm**.

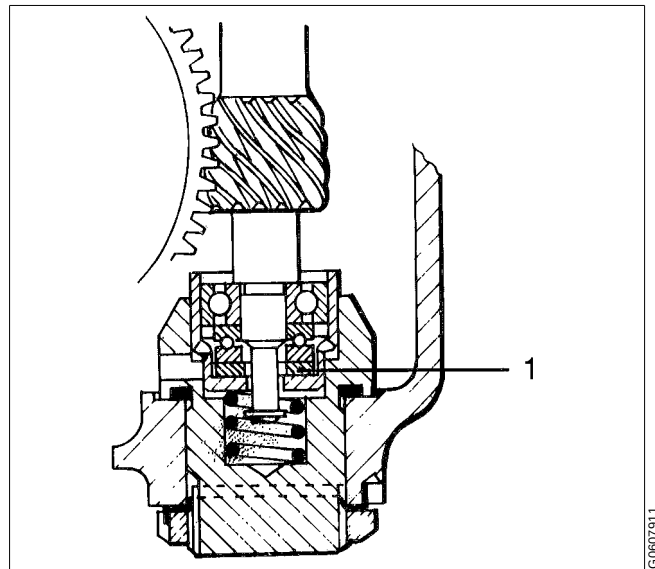
Adjustment is made with adjustment rings (1) under the thrust bearing at the bottom of the spindle.



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G0607811

5.3.8 Bowl spindle cone and bowl body nave

Impact marks on the spindle cone or in the bowl body nave may cause poor fit and out-of-balance vibrations.

The bowl spindle and the nave should also be checked if the bowl spindle has been dismantled or if the bowl runs roughly.

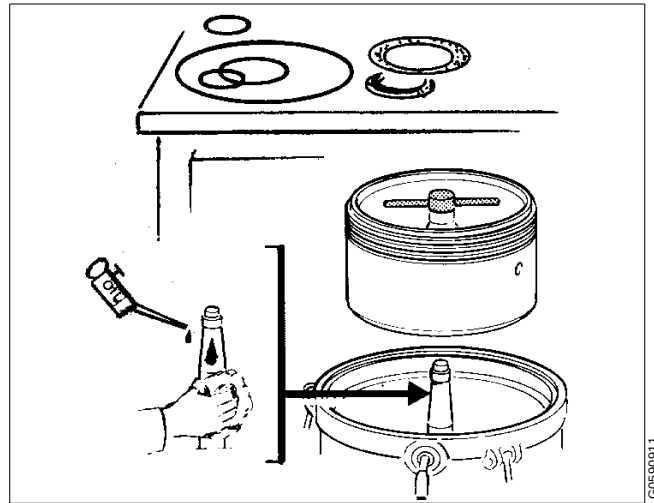
Corrosion may cause the bowl to stick firmly to the spindle cone and cause difficulties during the next dismantling.

- Remove any impact marks with a scraper and/or whetstone.

Rust can be removed by using a fine-grain emery cloth (e.g. No 320). Finish with polishing paper (e.g. No 600)

NOTE

Always use a scraper with great care. The conicity must not be marred.



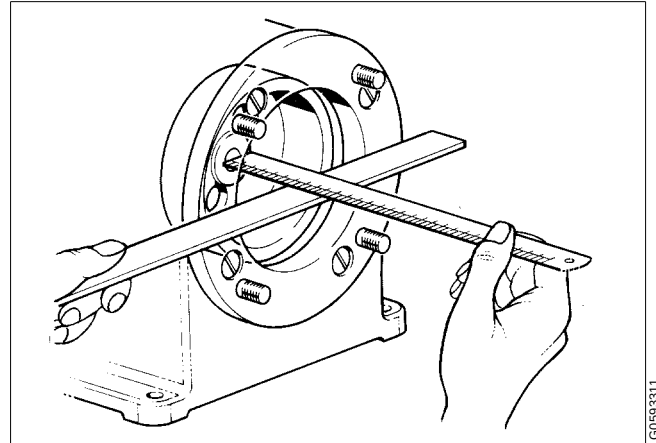
Put a little oil the bowl spindle to prevent corrosion

5.3.9 Coupling disc of motor

The position of the coupling disc on the motor shaft is establishing the correct location to the flexible coupling.

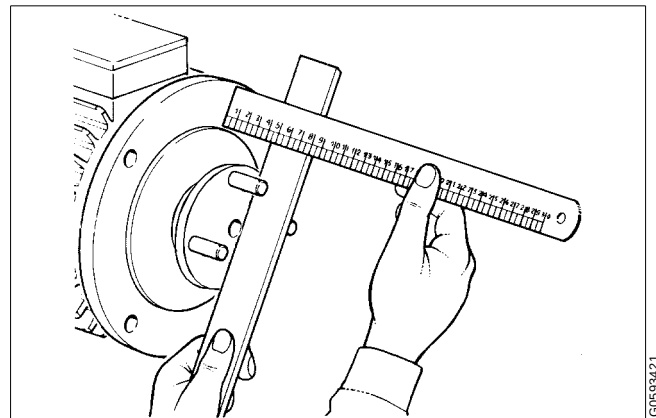
If the coupling disc is loosened without first marking its position on the motor shaft, the correct position must be determined again.

1. Measure the distance on the frame.



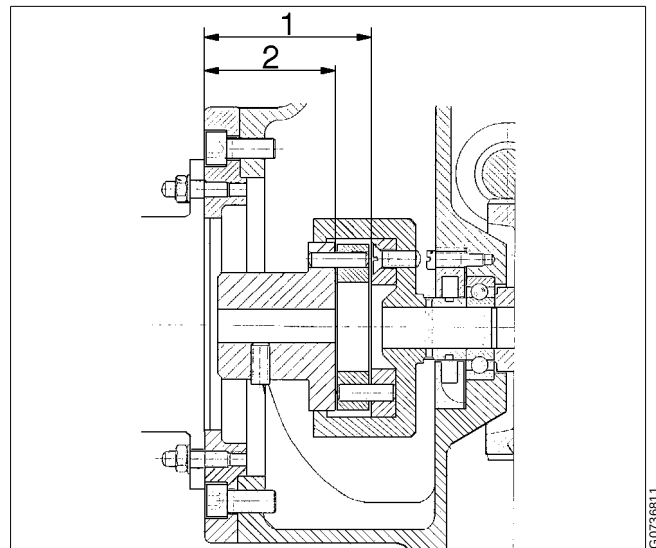
Frame distance

2. Measure the distance on the motor.



Motor distance

3. The coupling disc is in correct position when frame distance (1) is 22 ± 1 mm larger than motor distance (2). See fig.



Position of coupling disc

5.3.10 Top bearing springs

Weakened or broken buffer springs (2) may give rise to machine vibration (rough bowl running).

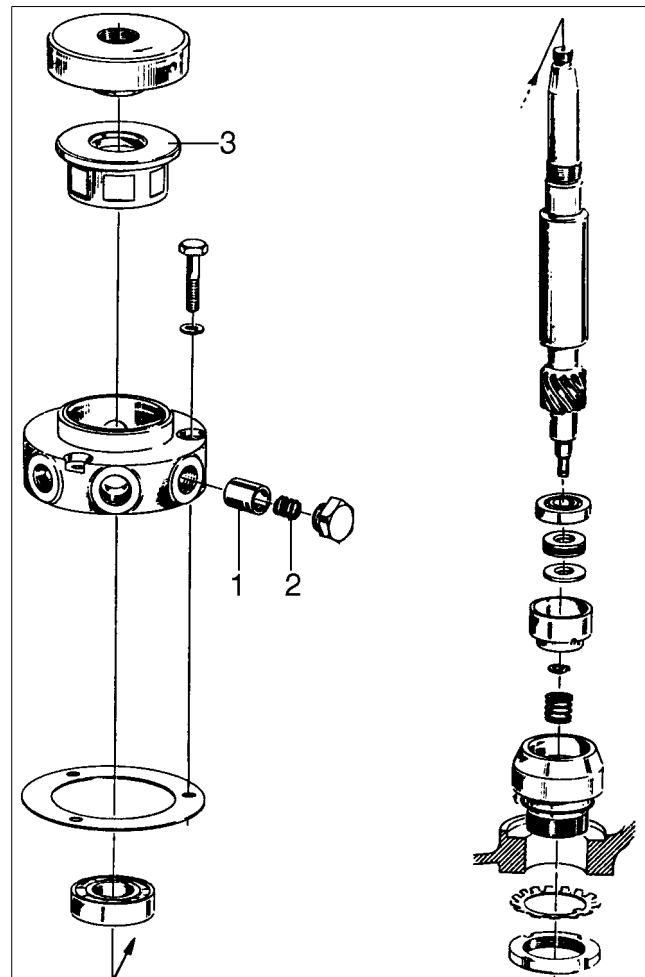
The condition (stiffness) of a spring can hardly be determined without using special testing equipment. So, an estimation of the spring condition must be based on the knowledge of the machine run before the overhaul. It is recommended, however, to replace all the springs at the annual overhaul.

In case of a sudden spring fracture, all springs should be replaced even when only one spring has broken.

5.3.11 Ball bearing housing

- Defective contact surfaces for the buffers on the ball bearing housing may give rise to machine vibration (rough bowl running).

Examine the contact surface for the buffers (1) on the ball bearing housing (3). In case of defects (indentations deeper than 0,1 mm) replace the housing as well as buffers and springs.



1. Radial buffer
2. Buffer spring
3. Ball bearing housing

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5.3.12 Worm wheel and worm; wear of teeth

Check the teeth of worm wheel and worm for wear.

See “5.3.13 Tooth appearance examples” on page 70.

Examine the contact surfaces and compare the tooth profiles. The gear may operate satisfactorily even when worn to some degree.

NOTE

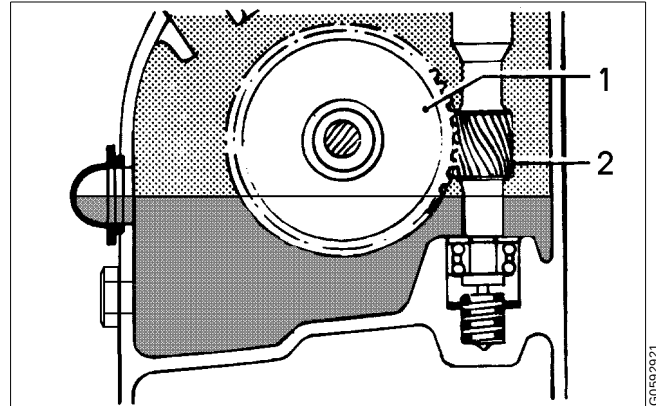
Replace both worm wheel and worm at the same time, even if only one of them is worn.

NOTE

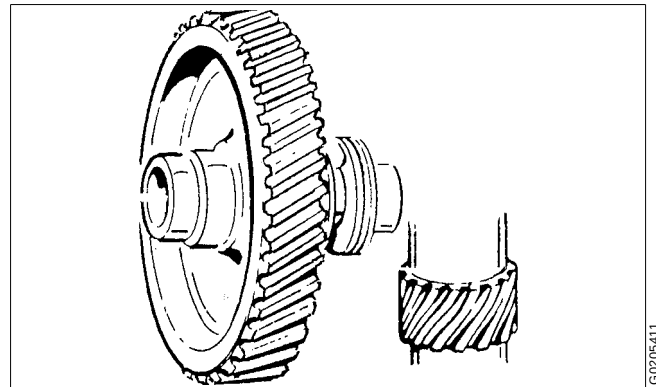
Presence of metal chips in the oil bath is an indication that the gear is wearing abnormally.

When using mineral-type oil in the worm gear housing, the presence of black deposits on the spindle parts is an indication that the oil base has deteriorated seriously or that some of the oil additives have precipitated. If pits are found on the worm gear, the cause could be that the additives are not suitable for this purpose.

In all these cases it is imperative to change to a high-temperature oil. See chapter “8.8 Lubricants” on page 137.



1. Worm wheel
2. Worm (part of bowl spindle)

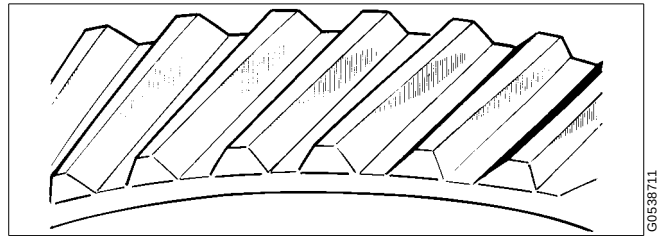


5.3.13 Tooth appearance examples

Satisfactory teeth

Uniform wear of contact surfaces. Surfaces are smooth.

Good contact surfaces will form on the teeth when the gear is subjected to only moderate load during its running-in period.

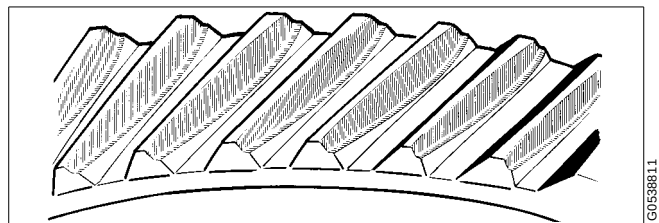


Satisfactory teeth

Worn teeth

Permissible wear is as a rule 1/3 of the thickness of the upper part of a tooth, provided that

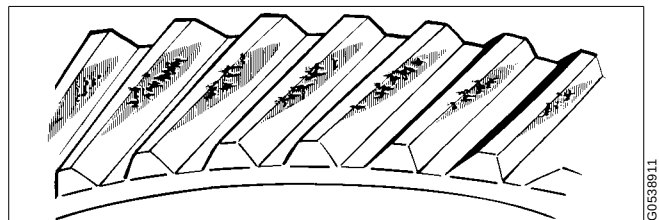
- the wear is uniform over the whole of the flank of a tooth
- and all teeth are worn in the same way.



Worn teeth

Spalling

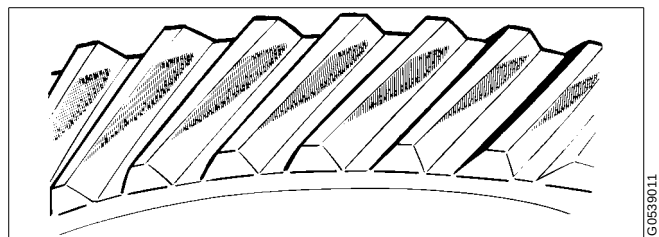
Small bits of the teeth have broken off, so-called spalling. This is generally due to excessive load or improper lubrication. Damage of this type need not necessitate immediate replacement, but careful checking at short intervals is imperative.



Spalling

Pitting

Small cavities in the teeth, so-called pitting, can occur through excessive load or improper lubrication. Damage of this type need not necessitate immediate replacement, but careful check at short intervals is imperative.



Pitting

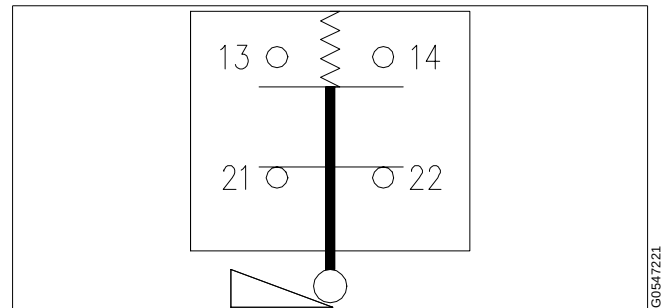
5.3.14 Cover interlocking switch, (Option)

Alfa Laval ref. 557822, rev. 1

Switch rating, voltage: Max. 500 V

current: Max. 3 A

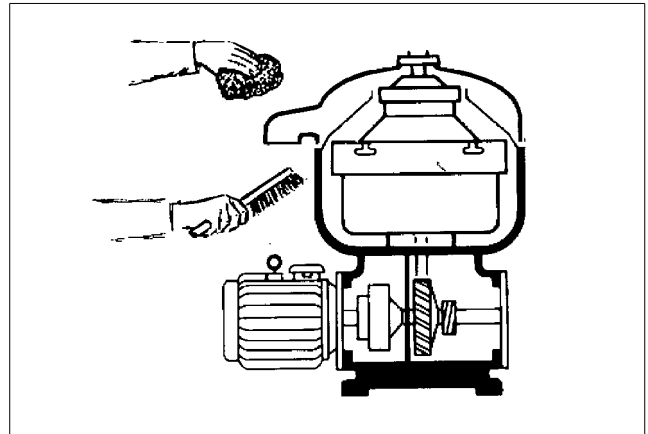
For other technical information see chapter “8.4 Connection list” on page 131 and chapter “8.7 Interface description” on page 134.



5.4 Cleaning

5.4.1 External cleaning

The external cleaning of the frame and motor should be restricted to brushing, sponging or wiping while the motor is running or is still hot.

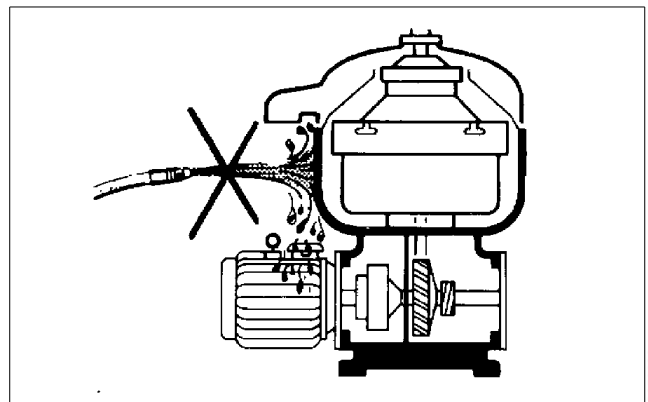


Use a sponge or cloth and a brush when cleaning

Never wash down a separator with a direct water stream.

Totally enclosed motors can be damaged by direct hosing to the same extent as open motors and even more than those, because:

- Some operators believe that these motors are sealed, and normally they are not.
- A water jet played on these motors will produce an internal vacuum, which will suck the water between the metal-to-metal contact surfaces into the windings, and this water cannot escape.
- Water directed on a hot motor may cause condensation resulting in short-circuiting and internal corrosion.



Never wash down a separator with a direct water stream or playing a water jet on the motor.

Be careful even when the motor is equipped with a protecting hood. Never play a water jet on the ventilation grill of the hood.

5.4.2 Cleaning agents

When using chemical cleaning agents, make sure you follow the general rules and supplier's recommendations regarding ventilation, protection of personnel, etc.

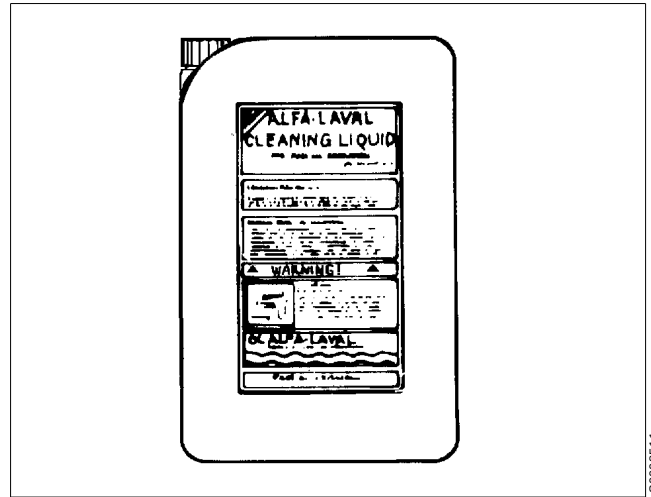
For separator bowl, inlet and outlet

A chemical cleaning agent must dissolve the deposits quickly without attacking the material of the separator parts.

- For cleaning of **lube oil** separators the most important function of the cleaning agent is to be a good solvent for the gypsum in the sludge. It should also act as a dispersant and emulsifier for oil. It is recommended to use **Alfa Laval cleaning liquid for lube oil separators** which has the above mentioned qualities. Note that carbon steel parts can be damaged by the cleaning agent if submerged for a long time.
- **Fuel oil** sludge mainly consists of complex organic substances such as asphaltenes. The most important property of a cleaning liquid for the removal of fuel oil sludge is the ability to dissolve these asphaltenes.

Alfa Laval cleaning liquid of fuel oil separators has been developed for this purpose. The liquid is water soluble, non-flammable and does not cause corrosion of brass and steel. It is also gentle to rubber and nylon gaskets in the separator bowl.

Before use, dilute the liquid with water to a concentration of 3-5%. Recommended cleaning temperature is 50-70 °C.



Alfa Laval cleaning liquid for lube and fuel oil separators is available in 25-litres plastic containers



CAUTION

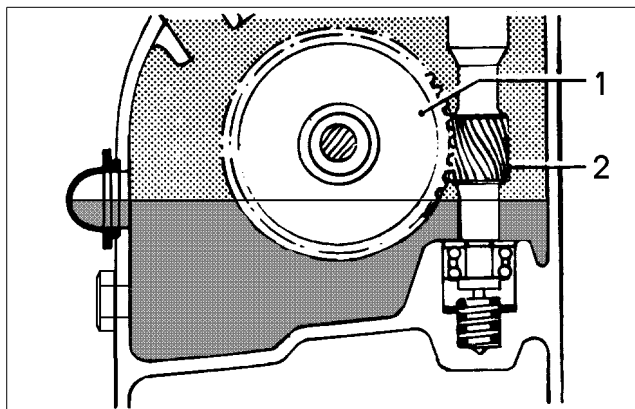
Skin irritation hazard

Read the instructions on the label of the plastic container before using the cleaning liquid.

Always wear safety goggles, gloves and protective clothing as the liquid is alkaline and dangerous to skin and eyes.

For parts of the driving devices

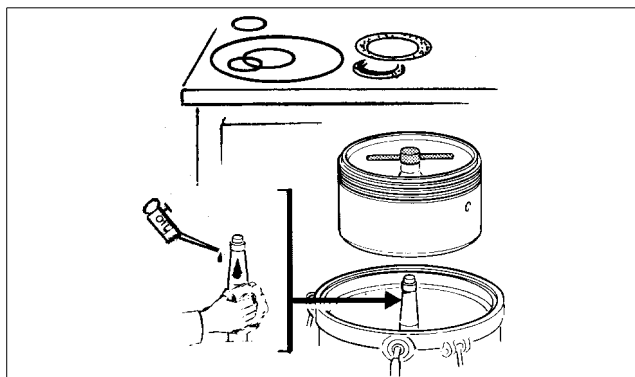
Use white spirit, cleaning-grade kerosene or diesel oil.



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Oiling (protect surfaces against corrosion)

Protect cleaned carbon steel parts against corrosion by oiling. Separator parts that are not assembled after cleaning must be wiped and coated with a thin layer of clean oil and protected from dust and dirt.



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5.4.3 Cleaning of bowl discs

Bowl discs

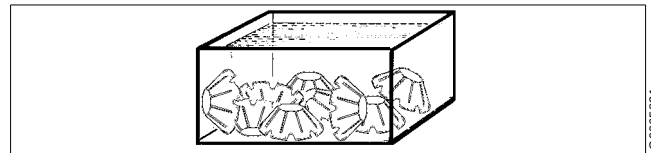
Handle the bowl discs carefully so as to avoid damage to the surfaces during cleaning.

NOTE

Mechanical cleaning is likely to scratch the disc surfaces causing deposits to form quicker and adhere more firmly.

A gentle chemical cleaning is therefore preferable to mechanical cleaning.

1. Remove the bowl discs from the distributor and lay them down, **one by one**, in the cleaning agent.
2. Let the discs remain in the cleaning agent until the deposits have been dissolved. This will normally take between two and four hours.
3. Finally clean the discs with a **soft** brush.



Put the discs one by one into the cleaning agent

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CAUTION

Cut hazard

The discs have sharp edges.



Clean the discs with a soft brush

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5.5 When changing oil

Check at each oil change

Check the teeth of both the worm wheel and worm for wear.

5.5.1 Oil change procedure

NOTE

Before adding or renewing lubricating oil in the worm gear housing, the information concerning different oil groups, handling of oils, oil change intervals etc. given in chapter 8, Technical Reference must be well known.

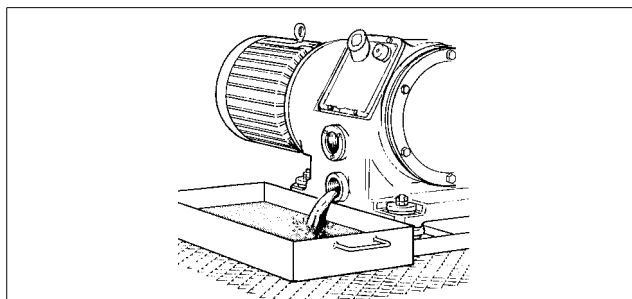
1. Place a collecting tray under the drain hole, remove the drain plug and drain off the oil.



CAUTION

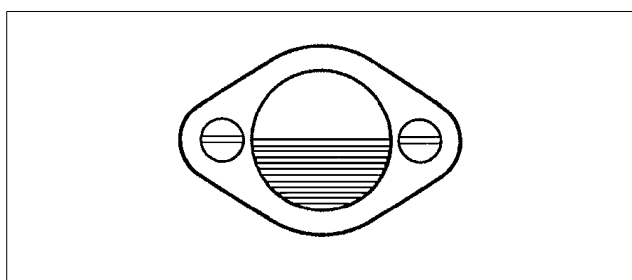
Burn hazards

Lubricating oil and various machine surfaces can be sufficiently hot to cause burns.



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2. Fill new oil in the worm gear housing. The oil level should be slightly above middle of the sight glass. See chapter "8.1 Technical data" on page 126.



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Check oil level

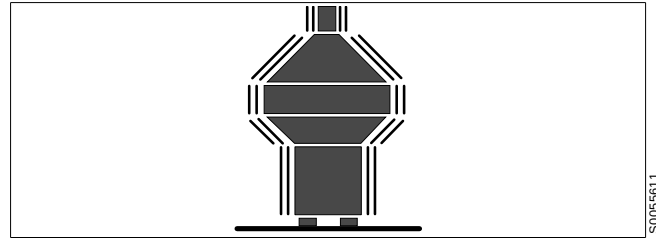
5.6 Common maintenance directions

5.6.1 Vibration

Vibration or noise often indicates that something is incorrect.

Use vibration analysis instrument to periodically check and record the level of vibration.

The level of vibration should not exceed 9,0 mm/s (RMS). Stop the separator and identify the cause.



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DANGER

Disintegration hazards

If excessive vibration occurs, **stop** separator and **keep bowl filled** with liquid during rundown.

The cause of the vibration must be identified and corrected before the separator is restarted. Excessive vibration can be due to incorrect assembly or poor cleaning of the bowl.

5.6.2 Ball and roller bearings

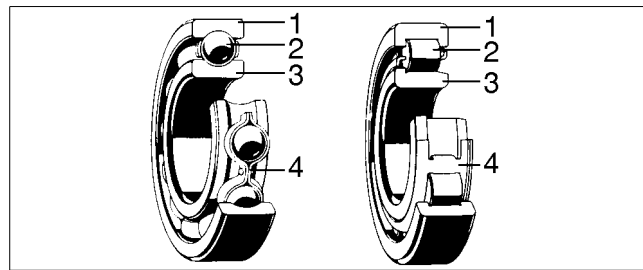
Use the greatest cleanliness when handling rolling bearings. Avoid unnecessary dismounting of bearings. Do not re-fit a used bearing. Always replaced it with a new one.

Important: Special design bearings for the bowl spindle.

The bearings used for the bowl spindle are specifically designed to withstand the speed, vibration, temperature and load characteristics of high-speed separators.

Do not use other bearings than those stated in the *Spare Parts Catalogue*.

A bearing that in appearance looks equivalent to the correct bearing may be considerably different from the latter in various respects: inside clearances, design and tolerances of the cage and ball (roller) races as well as material and heat treatment. Any deviation from the correct bearing may cause a serious breakdown.



1. Outer race
2. Ball / roller
3. Inner race
4. Cage

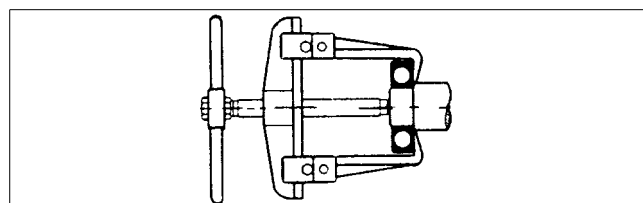
G0587321

Dismounting

Detach the bearing from its seat by pressing against the race having the tightest fit. Use a puller or a special tool. Thus, apply the pressure to the inner race when the bearing sits tightly on the shaft, and to the outer race when the bearing is tightly fitted in the housing respectively.

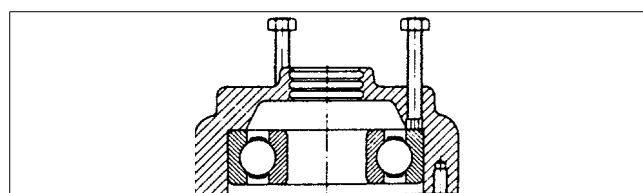
Arrange dismounted bearings and other parts in assembling order to avoid confusion.

Check the shaft end and the bearing seat in the housing for damage indicating that the bearing has rotated on the shaft, and in the housing respectively. Replace the damaged part, if the faults cannot be remedied by polishing or in some other way.



Use puller tool to dismantle bearings

G0590311

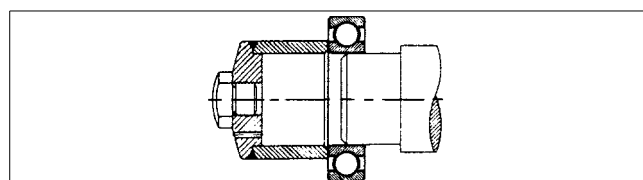


Check bearing seat

G0602211

Fitting

Leave new bearings in original wrapping until ready to fit. The anti-rust agent protecting a new bearing need not to be removed.



Use assembly tool

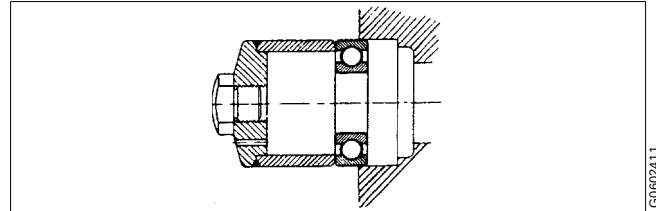
G0690411

Fit a bearing on a shaft by pressure applied to the inner race and in a housing by pressure applied to the outer race. Use a suitable piece of pipe or a metal drift and a hammer. Never hit the bearing directly.

Bearings sitting with tight fit on a shaft should be heated in oil before assembly. The oil temperature should not exceed 100 °C. Never leave the bearing in the oil bath longer than required for thorough heating.

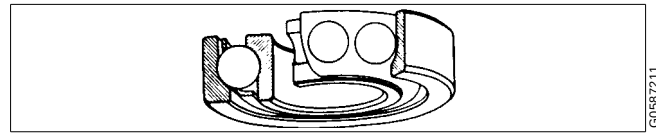
Angular contact ball bearings

Always fit single-row angular contact ball bearings with the stamped side of the inner race facing the axial load.



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Use assembly tool



G0587211

Angular contact bearing must be assembled correctly

5.6.3 Shutdowns

If the separator is shut down for a period of time, the following must be carried out:

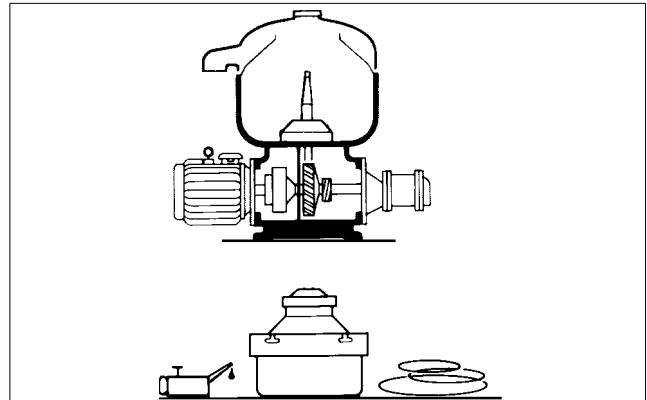
- Remove the bowl, according to instructions in chapter “6 Dismantling/Assembly” on page 83.

NOTE

The bowl must not be left on the spindle during standstill for more than one week.

Vibration in foundations can be transmitted to the bowl and produce one-sided loading of the bearings.

The resultant indentations in the ball bearing races can cause premature bearing failure.



Dismantling for shutdown

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- Protect cleaned carbon steel parts against corrosion by oiling. Separator parts that are not assembled after cleaning should be wiped and protected against dust and dirt.
- The O-rings should be removed.
- If the separator has been shut down for more than 12 months, a Major Service (MS) should be carried out.